

Set Theory

Labix

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Abstract

These notes aim to develop basic notions of sets and logics following closely to ZFC set theory, as well as introducing a few examples and provide proofs for theorems that require careful inspection. Theorems that with proofs omitted are expected to have readers be able to prove them. Although that there is a wide variety of axiomatic set theories that are accepted by different mathematicians, ZFC set theory is considered to be one of the most widely recognized theories among them all.

Beware that while set theory has no formal prerequisites in terms of its content (except perhaps logic theory), the mathematical concepts are not at all easy to understand. In fact the more foundational the mathematics, the harder the proofs and the more abstract the content becomes. First year university students should aim to complete up to chapter 3 in order to develop the suitable language for further mathematical content in their degrees.

Famous mathematicians who contributed to this area in mathematics include Ernst Zermelo and Abraham Fraenkel, Kurt Gödel and Georg Cantor and many more.

References

- Naive Set Theory by Paul R. Halmos

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Introduction to Formal Mathematics

Mathematics is built from three kinds of statements: **definitions**, **axioms**, and **theorems**. Understanding the difference between them is essential before reading any mathematical text.

Definitions

A **definition** introduces a new word or symbol and gives it a precise meaning. Definitions are neither true nor false — they are conventions. When we define a set to be a collection of objects, we are not making a claim about the world; we are deciding what the word *set* will mean for the rest of these notes.

A good definition does two things. It is **precise**: it draws a sharp boundary between what falls under the term and what does not. And it is **non-circular**: it does not define a term using that same term. Some notions, however, are so fundamental that they cannot be defined in terms of anything simpler. These are called **primitive notions** and are accepted as intuitively understood. In these notes, *set* and *membership* ($\in$) are primitive notions.

Axioms

An **axiom** is a statement we accept as true without proof. Axioms are the starting points of mathematics — the rules of the game. They are not arbitrary: a good axiom captures something so intuitively obvious, or so fundamental to the subject, that assuming it requires no justification.

It is important to understand what an axiom is *not*. An axiom is not a definition: it makes a genuine claim about mathematical objects rather than simply naming them. And it is not a theorem: it is not derived from anything more basic. Everything else in mathematics is ultimately built on top of the axioms.

In these notes the axioms govern what sets exist and how they behave. For example, the Axiom of Extensionality asserts that two sets with the same elements are equal. This is not something we prove — it is a fundamental assumption about the nature of sets that we decide to make at the outset.

Theorems

A **theorem** is a statement that has been proved to be true, starting from the axioms and definitions, using logical reasoning alone. A **proof** is the argument that establishes this. Every step in a proof must follow from the axioms, from previously proved theorems, or from the definitions, by the rules of logic.

Theorems go by various names depending on their importance and role:

- A **lemma** is a theorem proved primarily as a stepping stone toward a larger result.
- A **proposition** is a theorem of moderate importance or straightforward proof.
- A **corollary** is a theorem that follows quickly and easily from a theorem just proved.

These distinctions are matters of convention and emphasis, not of logical significance. A lemma is just as true as a theorem.

The Hierarchy

The relationship between these three kinds of statements is strictly ordered:

Primitive notions $\longrightarrow$ **Definitions** $\longrightarrow$ **Axioms** $\longrightarrow$ **Theorems**

Primitive notions are understood intuitively. Definitions give precise names to mathematical objects built from those primitives. Axioms make assumptions about those objects. Theorems derive consequences from those assumptions. Nothing flows in the other direction: a theorem cannot be used to justify an axiom, and a definition cannot make a claim that requires proof.

This hierarchy is what gives mathematics its distinctive character: once the axioms are fixed, the theorems are not a matter of opinion or experiment. They follow by pure logical necessity.

1 Basic Logic

Before we can talk about sets, we need a language for reasoning precisely. This section introduces the logical symbols and conventions for mathematical reasoning. The treatment is mainly informal since we are building intuition.

1.1 Logical Connectives

Definition 1.1.1 (Proposition)

A proposition is an assertion that is either true or false and not both.

Definition 1.1.2 (Negation)

Let P be a proposition. Define the negation of P to be the proposition $\neg P$ such that the following are true.

- $\neg P$ is true if P is false.
- $\neg P$ is false if P is true.

Definition 1.1.3 (Conjunction)

Let P, Q be propositions. Define the conjunction of P and Q to be the proposition $P \wedge Q$ such that the following are true.

- $P \wedge Q$ is true if P and Q are true.
- $P \wedge Q$ is false otherwise.

Definition 1.1.4 (Disjunction)

Let P, Q be propositions. Define the disjunction of P and Q to be the proposition $P \vee Q$ such that the following are true.

- $P \vee Q$ is true if P or Q is true.
- $P \vee Q$ is false otherwise.

Definition 1.1.5 (Conditional)

Let P, Q be propositions. Define the conditional of P and Q to be the proposition $P \rightarrow Q$ such that the following are true.

- $P \rightarrow Q$ is false if P is true and Q is false.
- $P \rightarrow Q$ is true otherwise.

Notice that $P \rightarrow Q$ is true whenever P is false. This is called vacuous truth. For example, the proposition “if the sky is purple, then the ocean is red” is vacuously true since the sky is not purple. In set theory, it is more common to use the symbol $\implies$.

Definition 1.1.6 (Biconditional)

Let P, Q be propositions. Define the biconditional of P and Q to be the proposition $P \leftrightarrow Q$ such that the following are true.

- $P \leftrightarrow Q$ is true if P and Q are true.
- $P \leftrightarrow Q$ is false otherwise.

In set theory, it is more common to use the symbol $\iff$.

Definition 1.1.7 (Compound Propositions)

A compound proposition is a proposition obtained by applying negation, conjunction, disjunction, conditional and biconditional to a collection of propositions.

Definition 1.1.8 (Tautologies and Contradictions)

Let P be a compound proposition.

- We say that P is a tautology if for any truth value of its proposition components, P is true.
- We say that P is a contradiction if for any truth value of its proposition components, P is false.

false.

1.2 Quantifiers

Definition 1.2.1 (Quantifiers)

The following symbols are called quantifiers.

- “ $\forall$ ” is the quantifier interpreted as “for all”.
- “ $\exists$ ” is the quantifier interpreted as “there exists”.

Definition 1.2.2 (Equality Symbol)

2 Introduction to Sets

The goal of Naive set theory is to introduce the concept of sets, and axioms of Zermelo–Fraenkel set theory together the axiom of choices (ZFC) and its consequences. Below is an overview of what each axiom means informally.

Axiom	Informal statement
Extensionality	Two sets are equal iff they have the same elements.
Empty Set	There exists a set with no elements.
Specification	Any definable sub-collection of a set is a set.
Replacement	The image of a set under a well-defined rule is a set.
Pairing	For any a, b , the set $\{a, b\}$ exists.
Union	For any collection $\mathcal{F}$, the union $\bigcup \mathcal{F}$ exists.
Regularity	Every non-empty set contains an element disjoint from it.
Power Set	For any set A , the collection of all subsets of A is a set.
Infinity	There exists an infinite set containing $\emptyset$ and closed under the successor operation.
Choice	Every collection of non-empty sets has a choice function.

2.1 Sets and Memberships

Definition 2.1.1 (Sets)

A set is a collection of objects called the elements of the set.

We take the word set as a primitive, undefined notion — just as geometry takes point and line as primitives. Intuitively, a set is a collection of objects, and those objects are called its elements or members.

Any attempt to define a set in terms of simpler concepts quickly runs into difficulty: what is a “collection”? What is an “object”? At some level, every mathematical theory must rest on notions that are understood intuitively rather than defined formally. For sets, that level is here. What we can do — and what the axioms below accomplish — is state precisely what sets are allowed to do, even if we cannot say precisely what they are.

Definition 2.1.2 (Membership)

Let A be a set.

- We write $x \in A$ to mean x is an element A .
- We write $x \notin A$ to mean that x is not an element of A .

2.2 The Axiom of Extensionality and the Axiom of Empty Set

Axiom 2.2.1 (The Axiom of Extensionality)

Let A, B be sets. Then A and B are equal if for any variable x , $x \in A$ if and only if $x \in B$. In first order logic symbols, we have

$$\forall A \forall B (A = B \leftrightarrow \forall t (t \in A \leftrightarrow t \in B))$$

Definition 2.2.2 (The Axiom of Empty Set)

There exists a set with no elements. In first order logic symbols, we have

$$\exists A \forall x (x \notin A)$$

Proposition 2.2.3

There exists a unique set with no elements.

Proof

The axiom of empty set provides the existence of such a set. The axiom of extensionality guarantees the set is unique. ■

Definition 2.2.4 (The Empty Set)

Define $\emptyset$ to be the set whose existence is given the axiom of empty set.

2.3 Subsets and the Axiom of Specification**Definition 2.3.1 (Subsets)**

Let A, B be sets.

- We say that $A \subseteq B$ if $x \in A \implies x \in B$.
- We say that $A \subset B$ if $x \in A \implies x \in B$ and $\exists b \in B, b \notin A$.

Proposition 2.3.2

Let A, B be sets. Then we have

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

Proof

Suppose first the $A = B$. Then for all $x \in A$, we have $x \in B$ by the axiom of extensionality. Hence $A \subseteq B$. Similarly, for $x \in B$ we have $x \in A$ by the axiom of extensionality. Hence $B \subseteq A$.

Conversely, suppose that $A \subseteq B$ and $B \subseteq A$. Then we know that $x \in A$ implies $x \in B$ and $y \in B$ implies $y \in A$. By the axiom of extensionality, we conclude that $A = B$. ■

Proposition 2.3.3

Let A, B, C be sets. Then the following are true.

- $\emptyset \subseteq A$.
- $A \subseteq A$.
- $A \subseteq B$ and $B \subseteq C$ implies $A \subseteq C$.

Axiom 2.3.4 (The Axiom of Specification)

Let A be a set. Let φ be a formula. Then there exists a set B such that $x \in B$ if and only if $x \in A$ and $\varphi(x)$. In first order logic symbols, we have

$$\forall A \exists B \forall x (x \in B \leftrightarrow (x \in A \wedge \varphi(x)))$$

Readers may verify that for any given set A and any formula φ , there a unique set B such that $x \in B$ if and only if $x \in A$ and $\varphi(x)$.

Definition 2.3.5 (Set Builder Notation for the Axiom of Specification)

Let A be a set. Let φ be a formula. Define

$$\{x \in A \mid \varphi(x)\}$$

to be the set whose existence is given the axiom of specification.

The axiom of specification requires the existence of an ambient set. This prevents a classical paradox in set theory.

Example 2.3.6 (Russell's Paradox)

Suppose we could form the set $R = \{x \mid x \notin x\}$.

- Suppose that $R \in R$. Then since it does not satisfy the defining condition $R \notin R$ of the set

R , we conclude that $R \notin R$.

- Suppose that $R \notin R$. Then since it does satisfy the defining condition $R \notin R$ of the set R , we conclude that $R \in R$.

Either way we reach a contradiction.

2.4 The Axiom of Replacement

Axiom 2.4.1 (The Axiom of Replacement)

Let A be a set. Let $\varphi(x, y)$ be a formula such that for all $x \in A$ there exists a unique y with $\varphi(x, y)$, there exists a set B such that $y \in B$ if and only if there exists $x \in A$ such that $\varphi(x, y)$. In first order logic symbols, we have

$$\forall A [\forall x \in A \exists! y \varphi(x, y)] \rightarrow \exists B \forall y [y \in B \leftrightarrow \exists x \in A \varphi(x, y)]$$

Definition 2.4.2 (Set Builder Notation for the Axiom of Replacement)

Let A be a set. Let $\varphi(x, y)$ be a formula such that for all $x \in A$ there exists a unique y with $\varphi(x, y)$, there exists a set B such that $y \in B$ if and only if there exists $x \in A$ such that $\varphi(x, y)$. Define

$$\{y \mid \exists x \in A \text{ such that } \varphi(x, y)\}$$

to be the set whose existence is given the axiom of replacement.

2.5 The Axiom of Pairing

Axiom 2.5.1 (The Axiom of Pairing)

For any objects a, b , there exists a set C such that for any variable x , $x \in C$ if and only if $x = a$ or $x = b$. In first order logic, we have

$$\forall a \forall b \exists C \forall x (x \in C \leftrightarrow (x = a \vee x = b))$$

Definition 2.5.2 (Set Builder Notation for the Axiom of Pairing)

Let a, b be objects. Define the unordered pair of a and b to be the set $\{a, b\}$ whose existence is given the axiom of pairing.

3 Operations on Sets

3.1 Unions and Intersections

Axiom 3.1.1 (The Axiom of Union)

Let A be a set. Then there exists a set B such that $x \in B$ if and only if there exists $C \in A$ such that $x \in C$. In first order logic symbols, we have

$$\forall A \exists B \forall x [x \in B \leftrightarrow \exists C (x \in C \wedge C \in A)]$$

Definition 3.1.2 (Union of Sets)

Let $\mathcal{F}$ be a set. Define the union of sets in $\mathcal{F}$ to be $\bigcup_{A \in \mathcal{F}} A$ whose existence is given the axiom of union.

Proposition 3.1.3

Let A, B, C be sets. Then the following are true.

- Identity: $A \cup \emptyset = A$
- Commutativity: $A \cup B = B \cup A$
- Associativity: $(A \cup B) \cup C = A \cup (B \cup C)$
- Idempotent: $A \cup A = A$
- $A \subset B \iff A \cup B = B$

Proof We prove it in order.

- $A \cup \emptyset = \{x : x \in A \text{ or } x \in \emptyset\} = \{x : x \in A\} = A$
- $A \cup B = \{x : x \in A \text{ or } x \in B\} = \{x : x \in B \text{ or } x \in A\} = B \cup A$
- Proved similarly by expanding the definition of union and using the fact that the logic operator "or" is commutative.
- $A \cup A = \{x : x \in A \text{ or } x \in A\} = \{x : x \in A\} = A$
- Suppose that $A \subset B$. $x \in A \cup B \implies x \in A \text{ or } x \in B \implies x \in B \text{ or } x \in B \implies x \in B$. Thus $A \cup B \subset B$. $x \in B \implies x \in B \text{ or } x \in B \implies x \in A \text{ or } x \in B \implies x \in A \cup B$. Thus $B \subset A \cup B$. Now suppose that $A \cup B = B$. $x \in A \implies x \in B$ thus $A \subset B$.

■

Definition 3.1.4 (Intersection of Sets)

Let $\mathcal{F}$ be a set. Let $C \in \mathcal{F}$ be a fixed object. Define the intersection of sets in $\mathcal{F}$ to be the set

$$\bigcap_{A \in \mathcal{F}} A = \{x \in C \mid \forall B \in \mathcal{F}, x \in B\}$$

given by the axiom of specification.

Proposition 3.1.5

Let A, B, C be sets. Then the following are true.

- Identity: $A \cap \emptyset = \emptyset$
- Commutativity: $A \cap B = B \cap A$
- Associativity: $(A \cap B) \cap C = A \cap (B \cap C)$
- Idempotent: $A \cap A = A$
- $A \subseteq B \iff A \cap B = A$

Proof

The first four are proved in similarity to the counterpart with unions. We prove the last item. Suppose that $A \subset B$. $x \in A \cap B \implies x \in A$ and $x \in B$. Thus $x \in A$ and $A \cap B \subset A$. $x \in A \implies x \in A$ and $x \in A \implies x \in A$ and $x \in B$. Thus $x \in A \cap B$ and $A \subset A \cap B$. Now

suppose that $A \cap B = A$. $x \in A \implies x \in A \cap B \implies x \in B$. Thus $A \subset B$. ■

Proposition 3.1.6 (Distributive Law)

Let A, B, C be sets. Then the following are true.

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Definition 3.1.7 (Disjoint Sets)

Let A, B be sets. We say that A and B are disjoint if and only if $A \cap B = \emptyset$.

3.2 The Axiom of Regularity

Axiom 3.2.1 (The Axiom of Regularity)

Let A be a non-empty set. Then there exists $x \in A$ such that $x \cap A = \emptyset$. In first order logic symbols, we have

$$\forall A [A \neq \emptyset \rightarrow (\exists x \in A)(x \cap A = \emptyset)]$$

Proposition 3.2.2

Let A be a set. Then we have $A \notin A$.

Proof

Apply the axiom of regularity to the set $\{A\}$. Then we have $A \cap \{A\} = \emptyset$. Hence $A \notin A$. ■

3.3 Set Complements

Definition 3.3.1 (Set Complements)

Let A, B be sets. Define the relative complement of B in A to be the set

$$A \setminus B = \{x \in A \mid x \notin B\}$$

given by the axiom of specification.

Proposition 3.3.2

Let E be a set. Let $A \subseteq E$ be a subset. Then the following are true.

- $(E \setminus A) \setminus A = A$.
- $E \setminus \emptyset = E$.
- $E \setminus E = \emptyset$.
- $A \cap E \setminus A = \emptyset$.
- $A \cup E \setminus A = E$.

Similar to the distributive law, we can associate the complement operator with the previous two operators, namely union and intersection.

Proposition 3.3.3 (De Morgans Laws)

Let E be a set. Let $A, B \subseteq E$ be subsets. Then the following are true.

- $E \setminus (A \cup B) = E \setminus A \cap E \setminus B$.
- $E \setminus (A \cap B) = E \setminus A \cup E \setminus B$.

Proposition 3.3.4

Let E be a set. Let $A, B \subseteq E$ be subsets. Then the following are true.

- $A \subseteq B$ if and only if $E \setminus B \subseteq E \setminus A$ if and only if $A \setminus B = \emptyset$.
- $A \setminus (A \setminus B) = A \cap B$.

3.4 The Axiom of Power Set

Axiom 3.4.1 (The Axiom of Power Set)

Let A be a set. Then there exists a set B such that $C \in B$ if and only if $C \subseteq A$. In first order logic symbols, we have

$$\forall A \exists B \forall C [C \in B \leftrightarrow \forall x (x \in C \rightarrow x \in A)]$$

Definition 3.4.2 (Power Sets)

Let A be a set. Define the power of set of A to be the set $\mathcal{P}(A)$ whose existence is given the axiom of power set.

Proposition 3.4.3

Let E be a set. Then the following are true.

- The power set distributes over intersections:

$$\bigcap_{X \in E} \mathcal{P}(X) = \mathcal{P}\left(\bigcap_{X \in E} X\right)$$

- The power set distributes over unions:

$$\bigcup_{X \in E} \mathcal{P}(X) \subseteq \mathcal{P}\left(\bigcup_{X \in E} X\right)$$

Proof

Let $A \in \bigcap_{X \in E} \mathcal{P}(X)$. Then $A \subseteq X$ for all $X \in E$. Thus $A \subseteq \bigcap_{X \in E} X$ and $A \in \mathcal{P}(\bigcap_{X \in E} X)$. For the reverse inclusion, the above implications are all double sided thus we are done.

Let $A \in \bigcup_{X \in E} \mathcal{P}(X)$. Then $A \subseteq \mathcal{P}(X)$ for some $X \in E$. Thus $A \subseteq \bigcup_{X \in E} X$ and $A \in \bigcup_{X \in E} \mathcal{P}(X)$ and we are done. Note that the reverse inclusion does not hold since we only have $A \subseteq \mathcal{P}(X)$ for some $X \in E$ as compared to the intersection of power sets. ■

3.5 Cartesian Products

Definition 3.5.1 (Ordered Pairs)

Let a, b be variables. Define the ordered pair of a and b to be the set

$$(a, b) = \{a, \{a, b\}\}$$

whose existence is given by the axiom of pairing.

Proposition 3.5.2

Let (a, b) and (c, d) be ordered pairs. Then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Proof

Suppose that $(a, b) = (c, d)$. Then we have by definition,

$$\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$$

There are two cases: $a = b$ and $a \neq b$. Suppose that $a = b$, then $\{\{a\}\} = \{\{c\}, \{c, d\}\}$. This forces $\{a\} = \{c\} = \{c, d\}$ and $a = c = d$. Suppose that $a \neq b$. Then $\{a\} = \{c\}$ and $\{a, b\} = \{c, d\}$. Thus $a = c$ and $b = d$. b cannot be c here since $a = b = c$ is a contradiction.

Now suppose that $a = c$ and $b = d$, then $\{a\} = \{c\}$ and $\{a, b\} = \{c, d\}$ thus $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$. ■

We now define the cartesian product of two sets, whose elements are ordered pairs.

Definition 3.5.3 (Cartesian Product)

Let A, B be sets. Define the Cartesian Product of A and B to be

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

Proposition 3.5.4 Let A, B, X, Y be sets.

- $(A \cup B) \times X = (A \times X) \cup (B \times X)$
- $(A \cap B) \times (X \cap Y) = (A \times X) \cap (B \times Y)$
- $(A \setminus B) \times X = (A \times X) \setminus (B \times X)$

Proof Again just a practise of expansion into logic. ■

4 Relations on a Set

4.1 Relations

Relations appear naturally in the course of studying mathematics. Some simple relations that are used commonly include $=$, $\leq$ and $\geq$ and $<$, $>$. In fact, $\subset$ and $\subseteq$ are also relations.

Definition 4.1.1 (Relation)

Let X, Y be sets. A relation from X to Y is a subset $R \subseteq X \times Y$.

Definition 4.1.2 (Types of Relations)

Let X, Y be sets. Let R be a relation on X .

- We say that R is reflexive if $(x, x) \in R$ for all $x \in X$
- We say that R is symmetric if $(x, y) \in R \implies (y, x) \in R$ for all $x, y \in X$
- We say that R is transitive if $(x, y) \in R$ and $(y, z) \in R$ implies $(x, z) \in R$ for all $x, y, z \in X$
- We say that R is anti-symmetric if $(x, y) \in R$ and $(y, x) \in R$ implies $x = y$.

Definition 4.1.3 (Equivalence Relation)

Let X be a set. Let R be a relation on X . We say that R is an equivalence relation if the following are true.

- R is reflexive
- R is symmetric
- R is transitive

Definition 4.1.4 (Inverse Relations)

Let X, Y be sets. Let R be a relation from X to Y . Define the inverse relation of R to be

$$R^{-1} = \{(b, a) \mid (a, b) \in R\} \subseteq Y \times X$$

4.2 Partitions and Equivalence Relations

Definition 4.2.1 (Partition)

Let X be a set. A partition of X is a subset $E \subseteq \mathcal{P}(X)$ such that the following are true.

- For any two sets $A, B \in E$ such that $A \neq B$, we have $A \cap B = \emptyset$ (A and B are disjoint).
- We have $\bigcup_{A \in E} A = X$.

Definition 4.2.2 (Equivalence Class)

Let X be a set. Let R be an equivalence relation on X .

- For each $x \in X$, define the equivalence class of x to be the set

$$[x] = \{y \in X \mid (x, y) \in R\}$$

- Define the set of all equivalence classes of X to be the set

$$X/R = \{[x] \mid x \in X\}$$

Theorem 4.2.3

Let X be a set. Then the following are true.

- Suppose that R is an equivalence relation on X . Then X/R is a partition of X .
- Suppose that E is a partition on X . Then the relation

$$R = \{(x, y) \in X \times X \mid \exists A \in E \text{ such that } x, y \in A\}$$

is an equivalence relation on X .

Proof

For every $x \in X$, $x \in x/R$ thus $\bigcup_{x \in X} x/R = X$. Now suppose that $z \in x/R \cap y/R$, then $(x, z) \in R$ and $(y, z) \in R$. By the symmetric property $(z, y) \in R$. By transitivity $(x, y) \in R$. Thus $x/R = y/R$. This proves that X/R is a partition.

We have that $(x, x) \in R$ for every x since they necessarily appear in the same set and none others (by definition of partition) thus the reflexive property holds. If $(x, y) \in R$ then x, y are in the same partition thus naturally $(y, x) \in R$ holds thus the symmetric property holds. Finally if $(x, y) \in R$ and $(y, z) \in R$ then x, y, z are all in the same partition thus $(x, z) \in R$ holds which proves transitivity. We can now conclude that R is an equivalence relation and we are done. ■

5 Functions between Sets

5.1 Functions

Functions play an integral role in all of mathematics. Therefore it is important to be able to express this concept in terms of set theoretic language.

Definition 5.1.1 (Functions)

Let X, Y be sets. A function from X to Y is a relation $f \subseteq X \times Y$ such that

- For all $x \in X$, there exists $y \in Y$ such that $(x, y) \in f$.
- For all $x \in X$, if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$.

In this case, we say that X is the domain of f and Y is the codomain of f . We write f as $f : X \rightarrow Y$ to indicate the domain and codomain of f .

Definition 5.1.2 (The Identity Function)

Let X be a set. Define the identity function of X to be the function $\text{id}_X : X \rightarrow X$ given by

$$\text{id}_X(x) = x$$

Definition 5.1.3 (The Inclusion Function)

Let Y be a set. Let $X \subseteq Y$ be a subset. Define the inclusion function from X to Y to be the function $\iota : X \rightarrow Y$ given by

$$\iota(x) = x$$

Definition 5.1.4 (The Restriction Function)

Let Y, Z be sets. Let $X \subseteq Y$ be a subset. Let $f : Y \rightarrow Z$ be a function. Define the restriction of f to X to be the function $f|_X : X \rightarrow Z$ given by

$$f|_X(x) = f(x)$$

Definition 5.1.5 (Bijective Functions)

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function.

- We say that f is injective if $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- We say that f is surjective if for all $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- We say that f is bijective if it is both injective and surjective.

This properties of a function does not only depend on how the function/relation is defined, but the domain and codomain also plays a part.

5.2 Composition of Functions

Definition 5.2.1 (Composition of Functions)

Let X, Y, Z be sets. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions. Define the composition of f and g to be the function $g \circ f : X \rightarrow Z$ given by

$$(g \circ f)(x) = g(f(x))$$

Proposition 5.2.2

Let X, Y, Z, W be sets. Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ and $h : Z \rightarrow W$ be functions. Then we have

$$(h \circ g) \circ f = h \circ (g \circ f)$$

In general, composition preserves injectivity and surjectivity, as seen by the following proposition.

Proposition 5.2.3

Let X, Y, Z be sets. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions. Then the following are true.

- If f and g are injective then $g \circ f$ is injective.
- If f and g are surjective then $g \circ f$ is surjective.
- If f and g are bijective then $g \circ f$ is bijective.
- If $g \circ f$ is injective, then f is injective.
- If $g \circ f$ is surjective, then g is surjective.

5.3 Inverse Functions

Inverse functions serve the important role of inverting functions so that we know what element in the domain is mapped to a fixed element in the codomain. However the condition that the inverse exists must be studied.

Definition 5.3.1 (Inverse Functions)

Let X, Y be sets. Let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ be functions. We say that g is an inverse of f if we have

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g = \text{id}_Y$$

In this case, we say that f is invertible.

Proposition 5.3.2

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function. If f is invertible, then the inverse relation f^{-1} is the unique function that is the inverse of f .

Theorem 5.3.3

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function. Then f is invertible if and only if f is bijective.

Proof We first suppose that the inverse of f is a function. We aim to prove injectivity and surjectivity.

Injectivity: Suppose that $f(x_1) = f(x_2) = y$. In terms of relations, this means that $(x_1, y) \in f$ and $(x_2, y) \in f$. By definition of inverse relations, $(y, x_1) \in f^{-1}$ and $(y, x_2) \in f^{-1}$. But since f^{-1} is a function, $x_1 = x_2$ and we are done.

Surjectivity: We want for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$. So choose an arbitrary $y \in Y$. Since f^{-1} is a function from Y to X , $f^{-1}(y) = x$ lies in X . Then $(y, x) \in f^{-1}$ implies that $(x, y) \in f$, which we are done.

Finally, suppose now that f is bijective. We aim to show that f^{-1} is a function. There are two items to show: that for every $y \in Y$, there exists $x \in X$ such that $f^{-1}(y) = x$ and that $(y, x) \in f^{-1}$ and $(y, z) \in f^{-1}$ implies $x = z$. For the first item, we use the fact that f is surjective. Let $y \in Y$. By surjectivity, there exists $x \in X$ such that $(x, y) \in f$. Then by definition of inverse relation $(y, x) \in f^{-1}$ and we are done.

Now for the second item, suppose that $(y, x) \in f^{-1}$ and $(y, z) \in f^{-1}$. Then by definition of inverse relation $(x, y) \in f$ and $(z, y) \in f$. Then using injectivity, we see that this should imply $x = z$ and so we are done. ■

Corollary 5.3.4

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function. If f is bijective, then f^{-1} is bijective.

Definition 5.3.5 (Images and Preimages)

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function.

- Let $A \subseteq X$ be a subset. Define the image of A under f to be the set

$$f(A) = \{f(x) : x \in A\}$$

- Let $B \subseteq Y$ be a subset. Define the preimage of B under f to be the set

$$f^{-1}(B) = \{x \in A \mid f(x) \in B\}$$

Proposition 5.3.6

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function. Let $A \subseteq X$ and $B \subseteq Y$ be subsets. Then the following are true.

- $f(f^{-1}(B)) \subseteq B$
- If f is surjective, then $f(f^{-1}(B)) = B$
- $A \subseteq f^{-1}(f(A))$
- If f is injective, then $f^{-1}(f(A)) = A$

Proof

Let $f : X \rightarrow Y$ be a function.

- Let $y \in f(f^{-1}(B))$. Then there exists $x \in f^{-1}(B)$ such that $y = f(x)$. But $x \in f^{-1}(B)$ implies $f(x) \in B$ by definition. Thus $y \in B$ and we are done.
- We just have to show that $B \subseteq f(f^{-1}(B))$. Let f be surjective and $y \in B$. Then by surjectivity there exists $x \in f^{-1}(B)$ such that $y = f(x)$. But $x \in f^{-1}(B)$ means that $f(x) \in f(f^{-1}(B))$. Thus $y \in f(f^{-1}(B))$ and we are done.
- Let $x \in A$. Then $f(x) \in f(A)$. But by definition of $f^{-1}(f(A))$, $x \in f^{-1}(f(A))$ thus we are done.
- We just have to show that $f^{-1}(f(A)) \subseteq A$. Let f be injective and $x \in f^{-1}(f(A))$. Then by definition, $f(x) \in f(A)$. By injectivity, there exists only one element in A that maps to $f(x)$, and that is precisely x . Thus $x \in A$ and we are done.

■

Proposition 5.3.7

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function. Let $A, B \subseteq Y$ be subsets. Then the following are true.

- $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$.
- $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$.
- $f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$.

Proposition 5.3.8

Let X, Y be sets. Let $f : X \rightarrow Y$ be a function. Let $A, B \subseteq X$ be subsets. Then the following are true.

- $f(A \cup B) = f(A) \cup f(B)$.
- $f(A \cap B) \subseteq f(A) \cap f(B)$.
- If f is injective, then $f(A \cap B) = f(A) \cap f(B)$.
- $f(X) \setminus f(A) \subseteq f(X \setminus A)$.
- If f is injective, then $f(X) \setminus f(A) = f(X \setminus A)$.

6 The Natural Numbers

6.1 The Axiom of Infinity

Before we formulate the natural numbers via set theoretic language, we need one more axiom to guarantee the existence of the set of natural numbers.

Definition 6.1.1 (Successor)

Let x be a set. Define the successor of x to be the set

$$x^+ = x \cup \{x\}$$

Axiom 6.1.2 (The Axiom of Infinity)

There exists a set containing the empty set and contains the successor of all its elements. In first order logic symbols, we have

$$\exists A(\emptyset \in A \wedge \forall x \in A, x^+ \in A)$$

6.2 The Peano The Axioms

Definition 6.2.1 (The Natural Numbers)

Define the set of natural numbers to be the smallest set containing $\emptyset$ and the successor of all its elements.

Theorem 6.2.2 (The Peano Axioms)

The following are true regarding the set $\mathbb{N}$.

- $\emptyset \in \mathbb{N}$.
- $n \in \mathbb{N}$ implies $n^+ \in \mathbb{N}$.
- $n^+ \neq \emptyset$ for all $n \in \mathbb{N}$.
- If $n, m \in \mathbb{N}$ and $n^+ = m^+$, then $n = m$.
- If $S \subseteq \mathbb{N}$ is a subset such that $\emptyset \in S$ and $n^+ \in S$ for all $n \in S$, then we have $S = \mathbb{N}$.

Thus each natural number is represented by the set of all smaller natural numbers. For instance, we have

$$0 = \emptyset, \quad 1 = \{\emptyset\}, \quad 2 = \{\emptyset, \{\emptyset\}\}$$

and so on.

Theorem 6.2.3 (Recursion Theorem)

Let X be a set. Let $a \in X$. Let $f : X \rightarrow X$ be a function. Then there exists a function $u : \mathbb{N} \rightarrow X$ such that the following are true.

- $u(0) = a$
- $u(n^+) = f(u(n))$ for all $n \in \omega$

Proof Let

$$E = \{A \subset \omega \times X \mid (0, a) \in A \text{ and } (n, x) \in A \implies (n^+, f(x)) \in A\}$$

Since $\omega \times X \in E$, $E \neq \emptyset$. Define $u = \bigcap_{B \in E} B$. Then $u \in E$. We want to show that u is a function. We prove that the set of all n such that $(n, x) \in u$ and $(n, y) \in u \implies x = y$ is ω . Let

$$S = \{n \in \omega \mid (n, x) \in u, (n, y) \in u \implies x = y\}$$

We prove it by the fifth item of Peano The Axiom.

We first show that $0 \in S$. Suppose for a contradiction that $0 \notin S$ and $(0, b) \in u$ and $a \neq b$. Consider $v = u - \{(0, b)\}$. $(0, a) \in v$ and $(n, x) \in A \implies (n^+, f(x)) \in A$ since $n^+ \neq 0$ for all $n \in \omega$. Thus $v \in E$ contradicts the fact that $u = \bigcap_{B \in E} B$.

We now show that $n^+ \in S$ if $n \in S$. Let $n \in S$. Then there exists a unique $x \in X$ such that $(n, x) \in u$. Suppose that $n^+ \notin S$. Then there exists $y \neq f(x)$ such that $(n, y) \in u$. Consider $v = u - \{(n, y)\}$. $(0, a) \in v$ and $(n, x) \in v \implies (n^+, f(x)) \in v$. Thus $v \in E$ contradicts the fact that $u = \bigcap_{B \in E} B$. We thus have $S = \omega$, finishing the proof. ■

Expanding things out, we see that $u(0) = a$, $u(1) = f(a)$, $u(2) = f(f(a))$ and so on. We have constructed a function u such that input the number in u means that you are composing that number of f to the initial element. This is why through this theorem, we allowed the existence of recursive functions. The application of this immediate as we will use it to define addition and multiplication through this theorem.

6.3 Arithmetic

We begin by defining the notion of addition in natural numbers.

Proposition 6.3.1

For every natural number $m \in \mathbb{N}$ there exists a function $s_m : \mathbb{N} \rightarrow \mathbb{N}$ such that

- $s_m(0) = m$
- $s_m(n^+) = (s_m(n))^+$

Proof

Set X as $\mathbb{N}$, f as the successor function in 6.2.2 and we are done. ■

The recursion here used is the repeated use of successors, meaning we are applying $+1$ a certain amount of times.

Definition 6.3.2 (Addition)

Define addition in $\mathbb{N}$ to be the function $+$: $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ given by

$$m + n = s_m(n)$$

We can now prove properties of addition in set theoretic language.

Proposition 6.3.3 (Properties of Addition)

Let $x, y, z \in \mathbb{N}$. Then the following are true regarding addition.

- Associative: $(x + y) + z = x + (y + z)$
- Existence of an Identity: $0 + x = x = x + 0$
- Commutativity: $x + y = y + x$

Proof

- For associativity, we induct on z . When $z = 0$, we have

$$(x + y) + 0 = x + y = x + (y + 0)$$

Suppose that $(x + y) + n = x + (y + n)$, then we have

$$\begin{aligned} (x + y) + n^+ &= ((x + y) + n)^+ \\ &= (x + (y + n))^+ && \text{(Induction Hypothesis)} \\ &= x + (y + n)^+ \\ &= x + (y + n^+) \end{aligned}$$

Thus by the principle of induction, we have associativity.

- For identity, we induct on x . When $x = 0$, we have that

$$0 + 0 = 0 = 0 + 0$$

Suppose that $0 + n = n = n + 0$, we have

$$\begin{aligned} 0 + n^+ &= (0 + n)^+ \\ &= n^+ && \text{(Induction Hypothesis)} \\ &= n^+ + 0 \end{aligned}$$

Thus by the principle of induction, we have $0 + x = x = x + 0$ for all $x \in \mathbb{N}$.

- For commutativity, we induct on y first to show that $x^+ + y = (x + y)^+$. When $y = 0$, we have $x^+ + 0 = x^+ = (x + 0)^+$. Now suppose that $x^+ + n = (x + n)^+$

$$\begin{aligned} x^+ + n^+ &= (x^+ + n)^+ \\ &= ((x + n)^+)^+ && \text{(Induction Hypothesis)} \\ &= (x + n^+)^+ \end{aligned}$$

Thus our first induction is complete. Now we prove commutativity by induction on x . When $x = 0$, we have $0 + y = y + 0$ from identity. Now suppose that $x + y = y + x$.

$$\begin{aligned} x^+ + y &= (x + y)^+ && \text{(First induction)} \\ &= (y + x)^+ && \text{(Induction Hypothesis)} \\ &= y + x^+ \end{aligned}$$

Thus by the principle of induction, we have commutativity. ■

We then proceed to multiplication.

Proposition 6.3.4

For every natural number $m \in \mathbb{N}$ there exists a function $p_m : \mathbb{N} \rightarrow \mathbb{N}$ such that the following are true.

- $p_m(0) = 0$
- $p_m(n^+) = p_m(n) + m$

Proof Take X as $\mathbb{N}$ and f as $f(x) = x + m$ for 6.2.2. ■

Similarly, the recursion here is repeated the repeated addition of $m \in \mathbb{N}$.

Definition 6.3.5 (Multiplication)

Define multiplication in $\mathbb{N}$ to be the function $\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ given by

$$m \cdot n = p_m(n)$$

Proposition 6.3.6 (Distributive Law of Natural Numbers)

Let $x, y, z \in \mathbb{N}$. Then we have

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

Proof

We prove the distributive law by induction on z . When $z = 0$, we have $x \cdot (y + 0) = x \cdot y$ and $x \cdot y + x \cdot 0 = x \cdot y$. Now suppose that $x \cdot (y + n) = x \cdot y + x \cdot n$.

$$\begin{aligned} x(y + n^+) &= x(y + n)^+ && \text{(Definition of Addition)} \\ &= x(y + n) + x && \text{(Definition of Multiplication)} \\ &= (xy + xn) + x && \text{(Induction Hypothesis)} \\ &= xy + xn + x && \text{(Associativity of Addition)} \\ xy + xn^+ &= xy + (xn + x) && \text{(Definition of Multiplication)} \\ &= xy + xn + x && \text{(Associativity of Addition)} \end{aligned}$$

Thus by the principle of induction, we have the distributive law. ■

Proposition 6.3.7

Let $x, y, z \in \mathbb{N}$. Then the following are true regarding multiplication.

- Associativity: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$
- Commutativity: $x \cdot y = y \cdot x$
- Existence of Identity: $1 \cdot x = x = x \cdot 1$

Proof

- We prove associativity by induction on z . When $z = 0$, $(x \cdot y) \cdot 0 = 0$ and $x \cdot (y \cdot 0) = x \cdot 0 = 0$. Now suppose that $(x \cdot y) \cdot n = x \cdot (y \cdot n)$. We have

$$\begin{aligned}
 (x \cdot y) \cdot n^+ &= (x \cdot y) \cdot n + x \cdot y && \text{(Definition of Multiplication)} \\
 &= x \cdot y + (x \cdot y) \cdot n && \text{(Commutativity of Addition)} \\
 &= x \cdot y + x \cdot (y \cdot n) && \text{(Induction Hypothesis)} \\
 &= x \cdot (y + y \cdot n) && \text{(Distributivity)} \\
 &= x \cdot (y \cdot n + y) && \text{(Commutativity of Addition)} \\
 &= x \cdot (y \cdot n^+) && \text{(Definition of Multiplication)}
 \end{aligned}$$

Thus by the principle of induction, we have the associative law.

- We prove commutativity by induction on y . When $y = 0$, we have $x \cdot 0 = 0 = 0 \cdot x$. Now suppose that $x \cdot y = y \cdot x$, we have

$$\begin{aligned}
 x \cdot y^+ &= (x \cdot y) + x && \text{(Definition of Multiplication)} \\
 &= y \cdot x + x && \text{(Induction Hypothesis)} \\
 &= x + \cdot (y + x) && \text{(Commutativity of Addition)} \\
 &= y^+ \cdot x && \text{(Distributivity)}
 \end{aligned}$$

Thus by the principle of induction, we have the commutative law.

- The identity is a special case of the commutative law, taking $y = 1$.



Now that we have the natural numbers, we can also extend the numbers into integers, rationals, real numbers and even complex numbers. Most of them without the use of set theoretic language. However, this is another topic unrelated to set theory mostly.

7 Ordering

7.1 Total and Partial Orders

Definition 7.1.1 (Partial Order)

Let X be a set. Let $\leq$ be a relation on X . We say that $\leq$ is a partial order on X if the following are true.

- $\leq$ is reflexive
- $\leq$ is anti-symmetric
- $\leq$ is transitive

In this case, we say that X is partially ordered, or that X is a poset.

Definition 7.1.2 (Strict Order)

Let X be a set. Let $\leq$ be a partial order on X . Define the strict order $<$ on X by $x < y$ if $x \leq y$ and $x \neq y$.

Definition 7.1.3 (Total Order)

Let X be a set. Let $\leq$ be a partial order on X . We say that $\leq$ is a total ordering on X if for any $x, y \in X$, either $x \leq y$ or $y \leq x$. In this case, we say that X is totally ordered.

Definition 7.1.4 (Least and Greatest Elements)

Let $(X, \leq)$ be a partially ordered set. Let $a \in X$.

- We say that a is a least element of X if $a \leq x$ for all $x \in X$.
- We say that a is a greatest element of X if $x \leq a$ for all $x \in X$.

Definition 7.1.5 (Minimal and Maximal Elements)

Let $(X, \leq)$ be a partially ordered set. Let $a \in X$.

- We say that a is a minimal element of X if $x \leq a$ implies $a = x$ for all $x \in X$.
- We say that a is a maximal element of X if $a \leq x$ implies $a = x$ for all $x \in X$.

Proposition 7.1.6

Let $(X, \leq)$ be a partially ordered set. Let $a \in X$. Then the following are true.

- If a is a least element of X , then a is a minimal element of X .
- If a is a greatest element of X , then a is a maximal element of X .

Proposition 7.1.7

Let $(X, \leq)$ be a totally ordered set. Let $a \in X$. Then the following are true.

- a is a least element of X if and only if a is a minimal element of X .
- a is a greatest element of X if and only if a is a maximal element of X .

7.2 Well-Ordered Sets

Definition 7.2.1 (Well-Ordered Sets)

Let X be a set. Let R be a total order on X . We say that X is well-ordered if any non-empty subset of X admits a least element.

Definition 7.2.2 (Initial Segments)

Let $(X, \leq)$ be a partially ordered set. Let $a \in X$. Define the initial segment of a to be the set

$$s(a) = \{x \in X \mid x < a\}$$

Theorem 7.2.3 (Principle of Transfinite Induction)

Let $(X, \leq)$ be a well-ordered set. Let $S \subseteq X$. Suppose that for all $x \in X$, we have $s(x) \subseteq S$ implies $x \in S$. Then we have $S = X$.

7.3 Ordering the Natural Numbers

Definition 7.3.1 (Strict Ordering on $\mathbb{N}$)

Define a relation $<$ on $\mathbb{N}$ as follows. Let $n, m \in \mathbb{N}$. Then $n < m$ if $n \in m$.

Theorem 7.3.2 (Trichotomy of the Ordering on $\mathbb{N}$)

Let $n, m \in \mathbb{N}$. Then exactly one of the following holds.

- $n < m$.
- $n = m$.
- $m < n$.

Definition 7.3.3 (Ordering in $\mathbb{N}$)

Define a relation $\leq$ on $\mathbb{N}$ as follows. Let $n, m \in \mathbb{N}$. Then $n \leq m$ if $n \in m$ or $n = m$.

Theorem 7.3.4 (Well-Ordering of the Natural Numbers)

$(\mathbb{N}, \leq)$ is a well-ordered set.

Corollary 7.3.5 (Principle of Strong Induction)

Let $S \subseteq \mathbb{N}$ be a subset. Suppose that for all $k \in \mathbb{N}$, we have $m < k$ and $m \in S$ implies $k \in S$. Then $S = \mathbb{N}$.

8 The Axiom of Choice and its Consequences

8.1 The Axiom of Choice

Axiom 8.1.1 (The Axiom of Choice)

Let $\mathcal{F}$ be a set such that $A \neq \emptyset$ for all $A \in \mathcal{F}$. Then there exists a function $c : \mathcal{F} \rightarrow \bigcup_{A \in \mathcal{F}} A$ such that $c(A) \in A$ for all $A \in \mathcal{F}$. In first order logic symbols, we have

$$\forall \mathcal{F} (\forall A \in \mathcal{F}, A \neq \emptyset) \rightarrow \exists c \forall A \in \mathcal{F}, c(A) \in A$$

8.2 Zorn's Lemma

Definition 8.2.1 (Chains)

Let $(X, \leq)$ be a partially ordered set. Let $C \subseteq X$ be a subset. We say that C is a chain if $(C, \leq)$ is totally ordered.

Theorem 8.2.2 (Zorn's Lemma)

Let $(X, \leq)$ be a partially ordered set. Suppose that every chain has a maximal element. Then X has a maximal element.

8.3 The Well-Ordering Theorem

Theorem 8.3.1 (The Well-Ordering Theorem)

Let X be a set. Then there exists a total order $\leq$ on X such that $(X, \leq)$ is well-ordered.

9 Cardinality and Infinity

9.1 Comparing Sizes of Sets

Definition 9.1.1 (Same Cardinality)

Let X, Y be sets. We say that X and Y have the same cardinality if there exists a bijective function $f : X \rightarrow Y$. In this case, we write $|X| = |Y|$.

Definition 9.1.2 (Cardinality Ordering)

Let X, Y be sets. We say that $|X| \leq |Y|$ if the following are true.

- There exists an injective function $f : X \rightarrow Y$.
- $|X| \neq |Y|$.

Theorem 9.1.3 (Cantor-Bernstein-Schroeder)

Let X, Y be sets. If $|X| \leq |Y|$ and $|Y| \leq |X|$, then we have $|X| = |Y|$.

Theorem 9.1.4 (Cantor's Theorem)

Let X be a set. Then we have $|X| < |\mathcal{P}(X)|$.

9.2 Countability

Definition 9.2.1 (Countability)

Let X be a set.

- We say that X is finite if $|X| = n$ for some $n \in \mathbb{N}$.
- We say that X is countably infinite if $|X| = |\mathbb{N}|$.
- We say that X is uncountable if $|\mathbb{N}| < |X|$.

Proposition 9.2.2

Let X, Y be finite sets. Then the following are true.

- If X and Y are disjoint, then $|X \cup Y| = |X| + |Y|$.
- $|X \times Y| = |X| \cdot |Y|$.

Proposition 9.2.3

Let $\mathcal{F}$ be a countable set. Suppose that all $A \in \mathcal{F}$ are countable. Then $\bigcup_{A \in \mathcal{F}} A$ is countable.

9.3 Ordinal Numbers